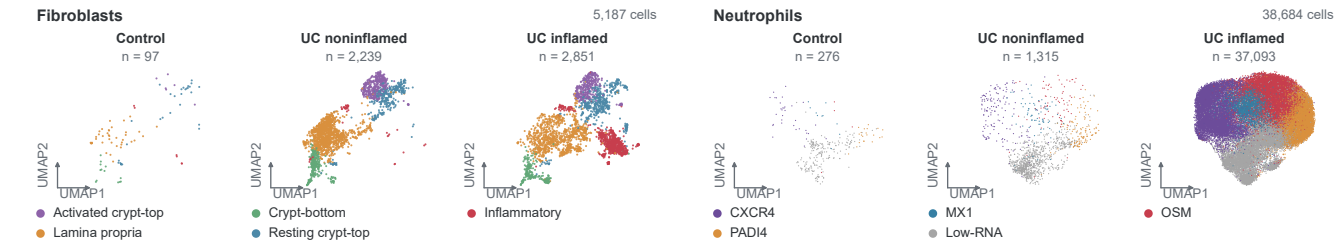
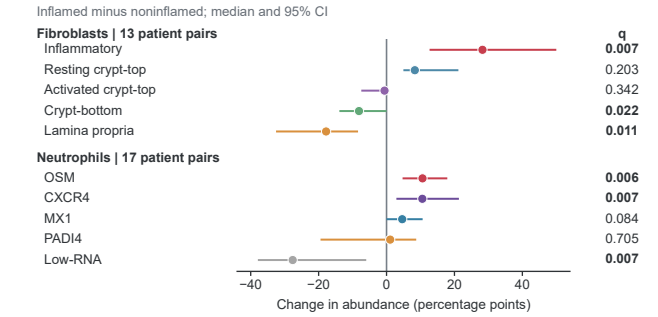


Figure 1

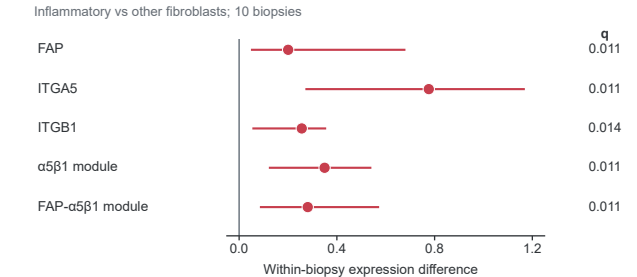
A Human UC fibroblast and neutrophil state landscape



B Paired cell-state remodeling

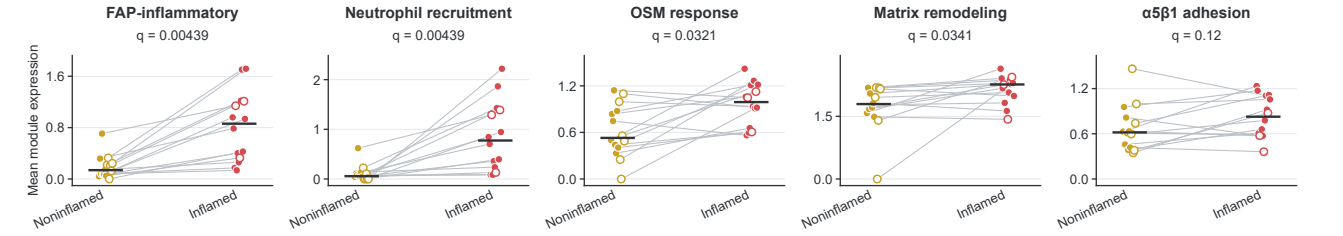


C Inflammatory-fibroblast activation



D Paired fibroblast program changes

14 patient pairs; lines connect patients; black bars, medians; open points, <10 recovered cells



E Neutrophil state-associated RNA programs

11 matched patients; ≥ 20 cells/state; naming genes excluded



F Predicted reciprocal signals

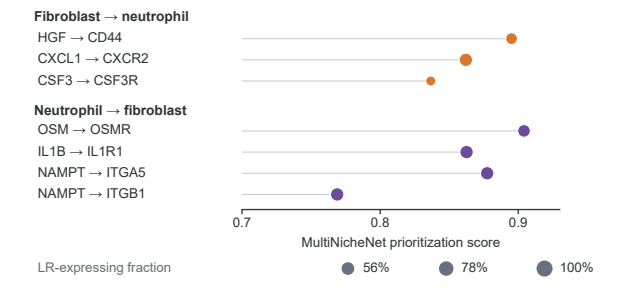
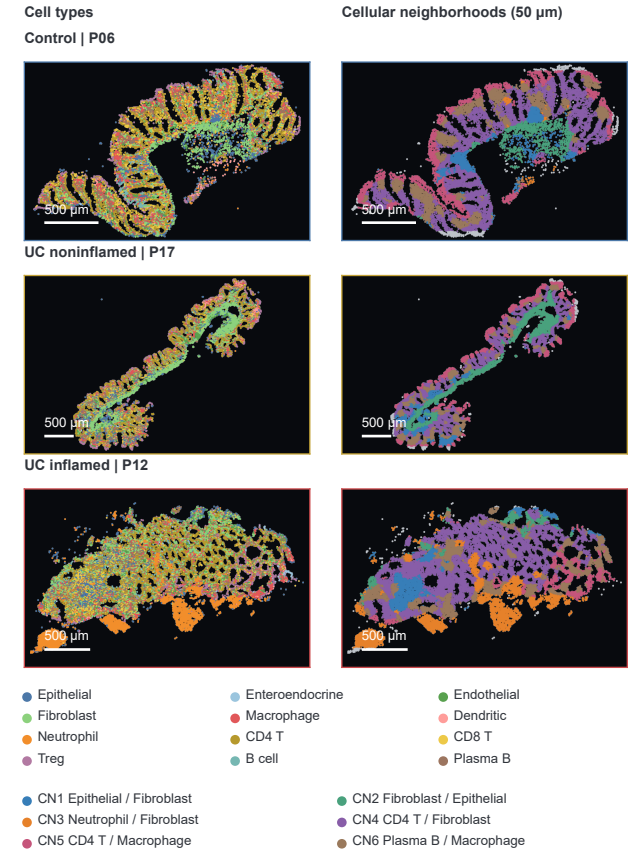


Figure 2

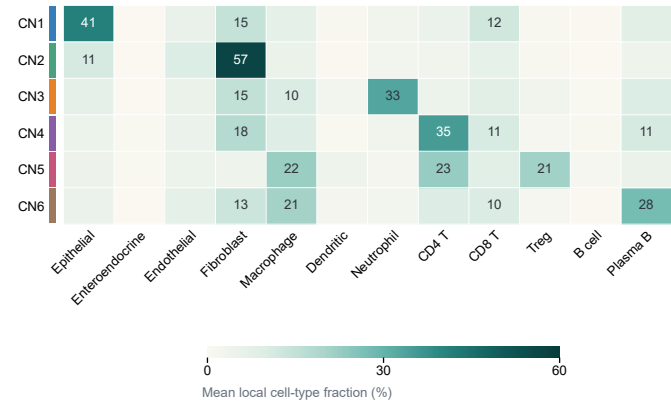
A CODEX cell types and neighborhoods

24 independent patients; matched fields and coordinates



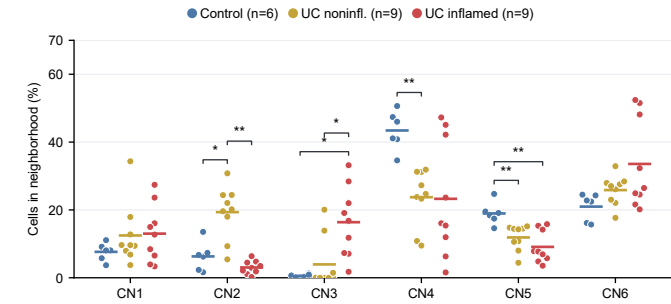
B Neighborhood composition

Patient-averaged local cell-type composition



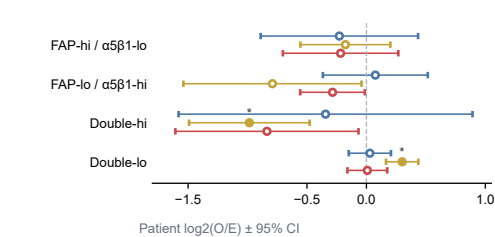
C Tissue neighborhood occupancy

Patient values and group means



D Availability-adjusted proximity

50- μ m proximity relative to a stratified spatial null



E Cell proximity and neutrophil markers

Adjusted associations in 8 patients with inflamed UC

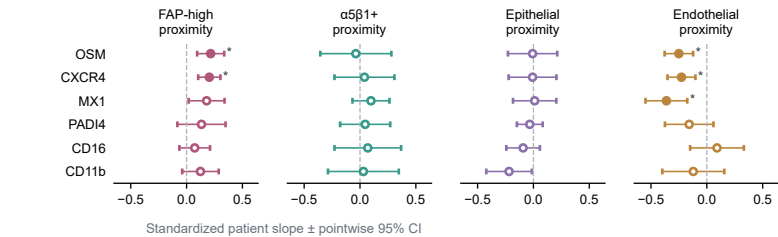
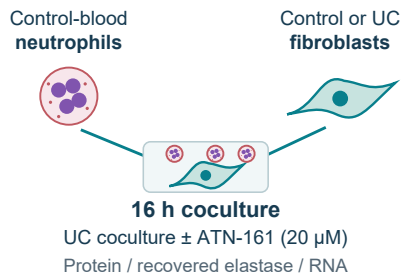
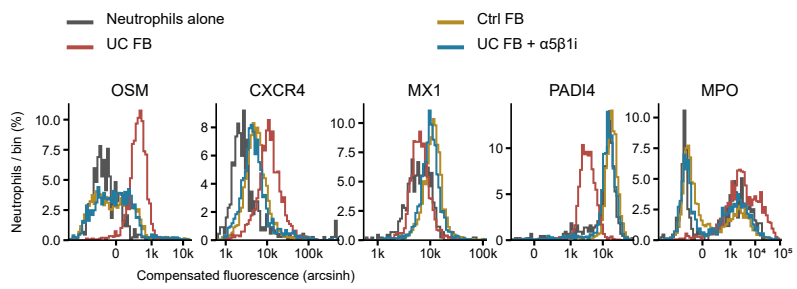


Figure 3

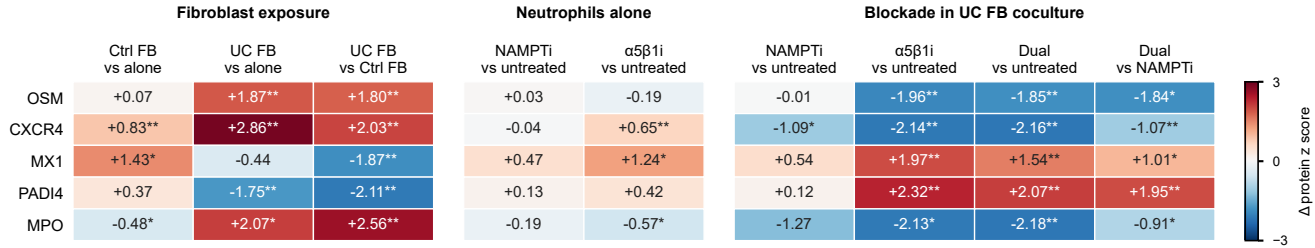
A Human coculture workflow



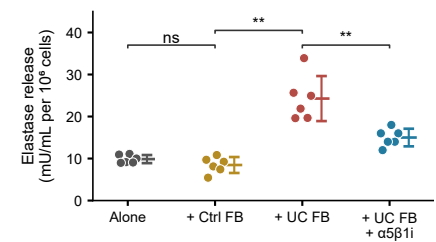
B Neutrophil protein distributions



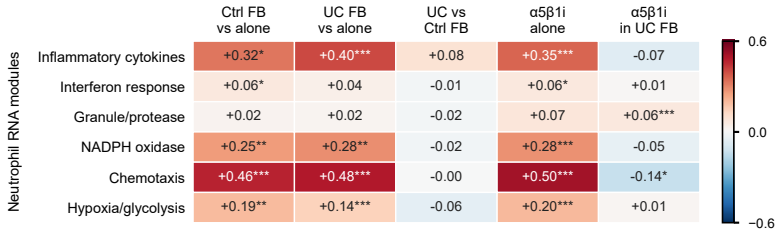
C Fibroblast exposure and blockade effects on neutrophil proteins



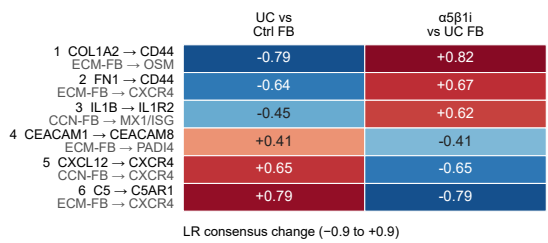
D NET-associated elastase activity



E Neutrophil RNA response programs



F Predicted fibroblast–neutrophil routes



G Module support for ligands in F

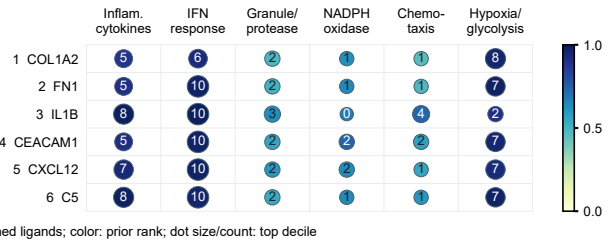
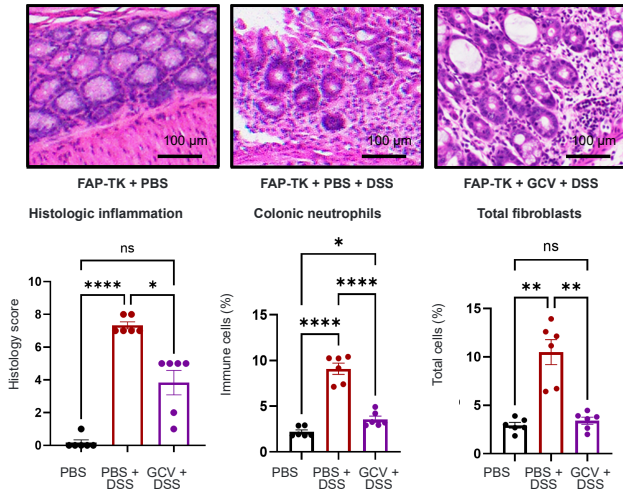
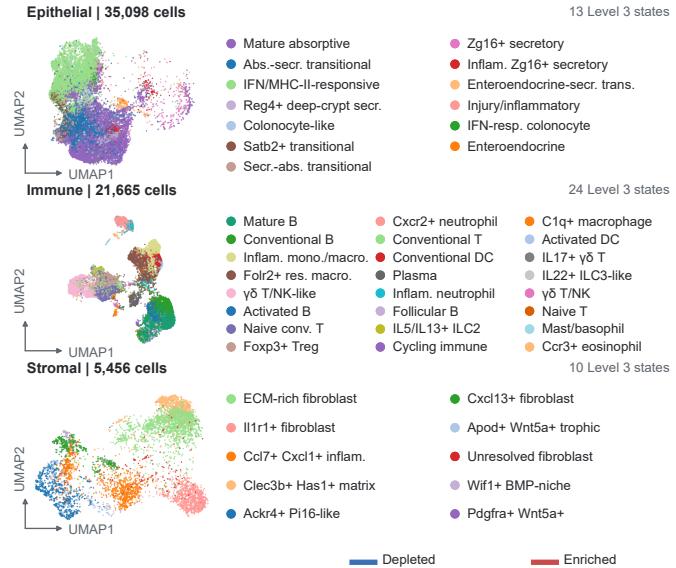


Figure 4

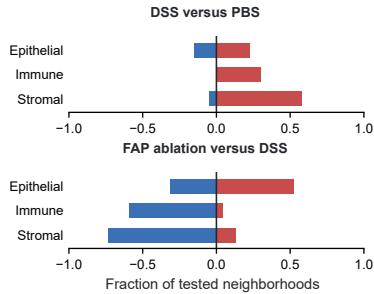
A FAP ablation reduces DSS-colitis pathology



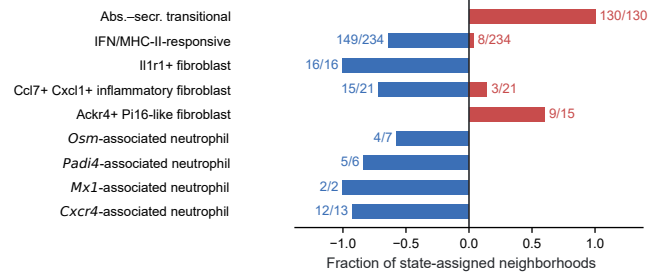
B Compartment-specific UMAPs



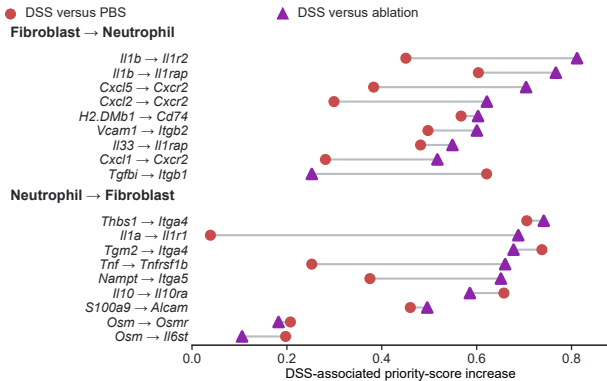
C MiloR neighborhood remodeling



Level 3 states | intervention versus DSS



D Reciprocal signalling routes



E Predicted receiver-cell pathways

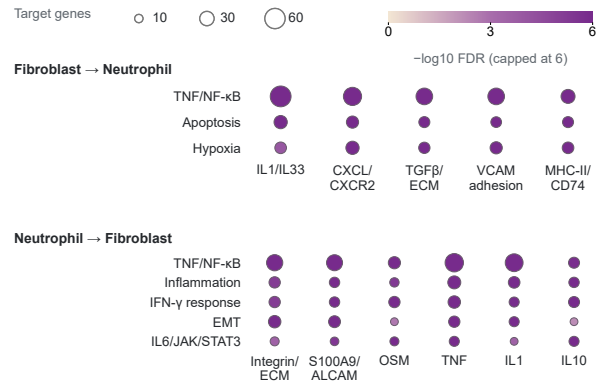
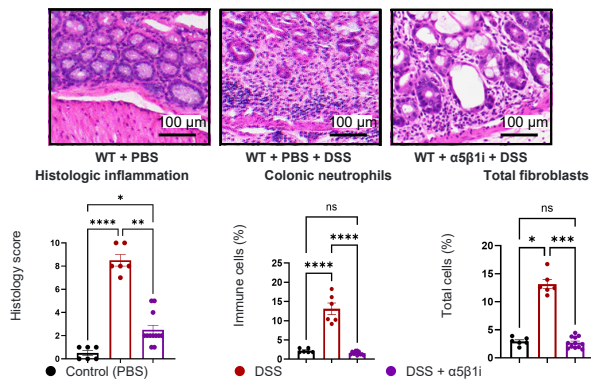
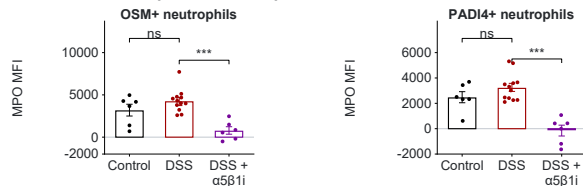


Figure 5

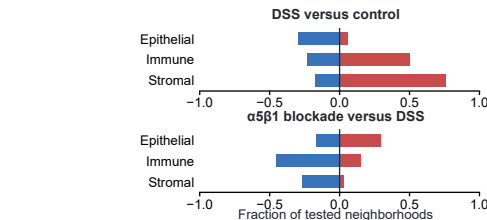
A α5β1 blockade reduces DSS-colitis pathology



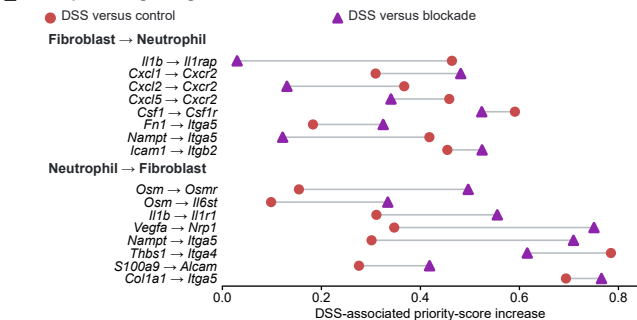
B MPO in marker-positive neutrophils



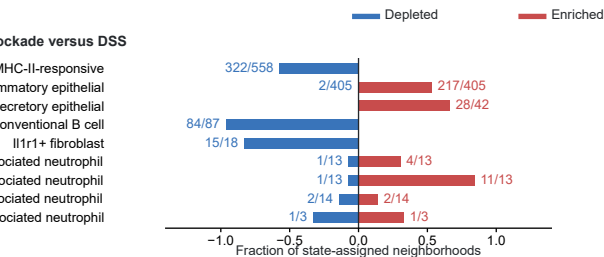
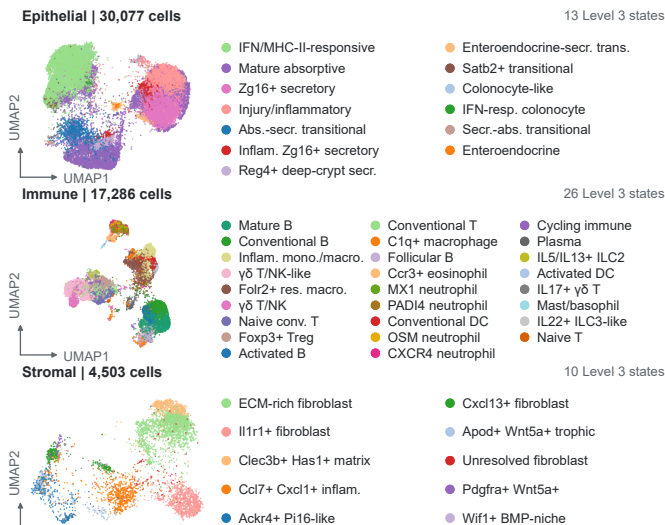
D Milor neighborhood remodeling



E Reciprocal signaling routes



C Compartment-specific UMAPs



F Predicted receiver-cell pathways

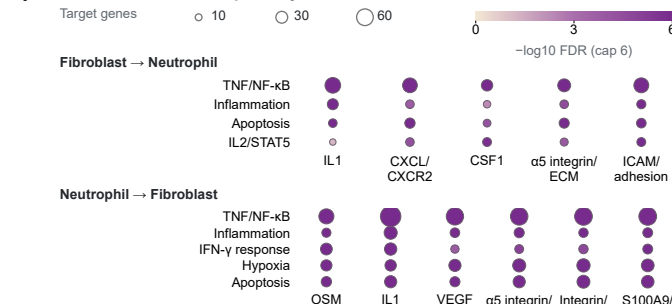
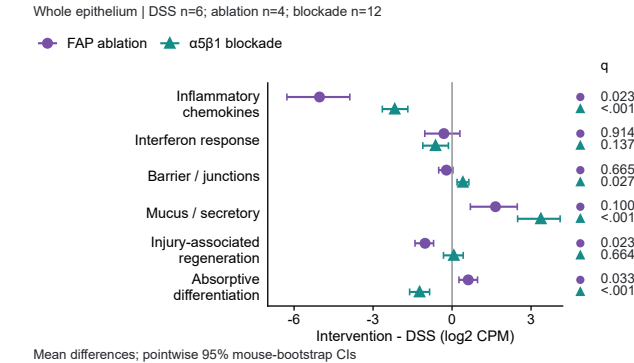
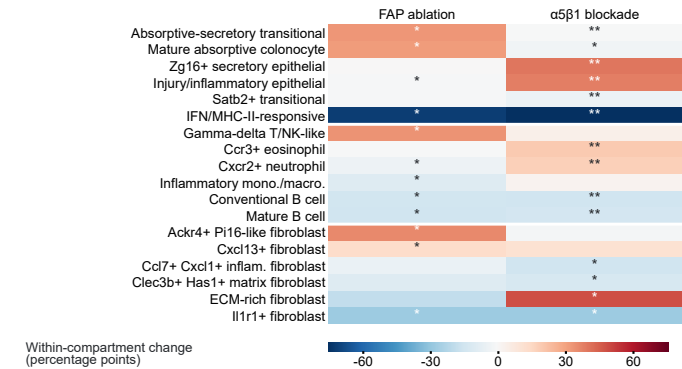


Figure 6

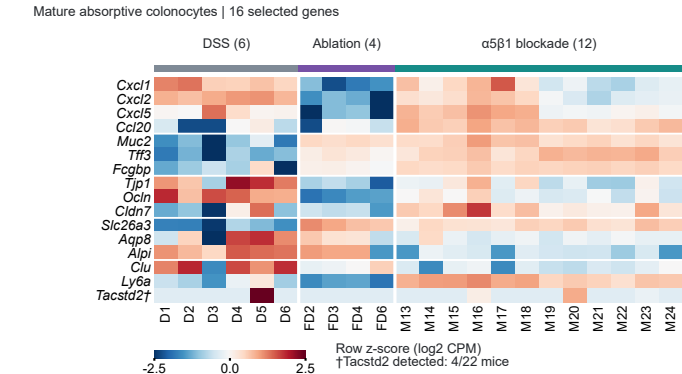
A Epithelial RNA programs



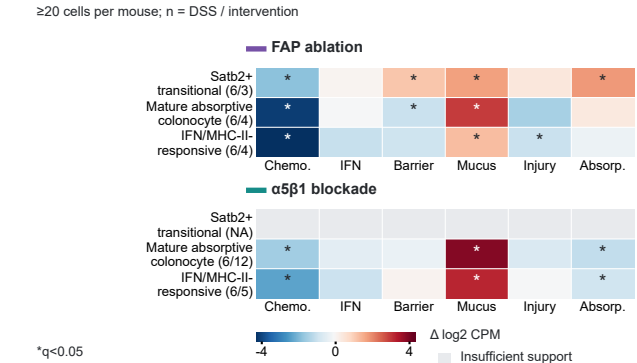
C Cell-state composition



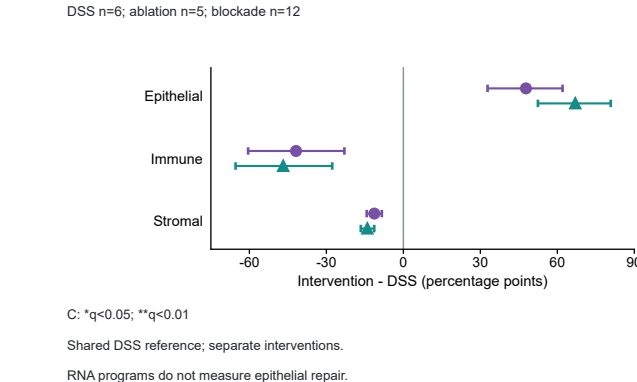
E Epithelial gene expression



B Within-state epithelial programs



D Recovered-cell proportions



F Individual-mouse epithelial profiles

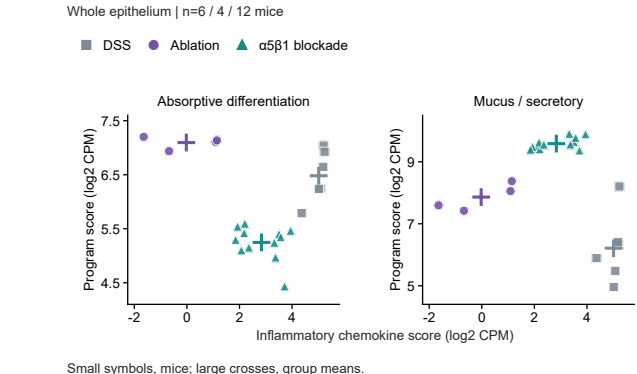
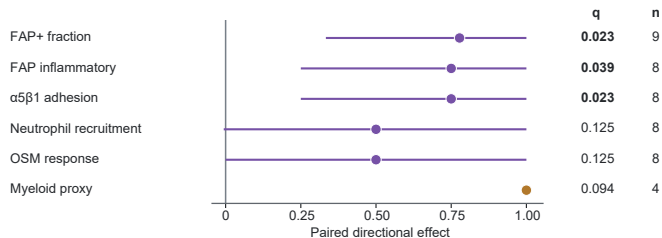


Figure 7

A TAURUS UC: paired inflammation

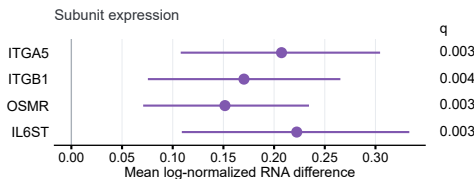
Inflamed vs noninflamed; 95% bootstrap CI



Myeloid: monocyte proxy; TAURUS did not retain neutrophils.

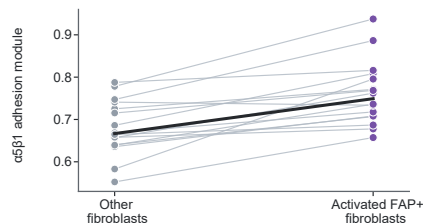
C Receptor components in activated FAP+ fibroblasts

Activated FAP+ minus other fibroblasts; 16 paired patients; 50 matched biopsies



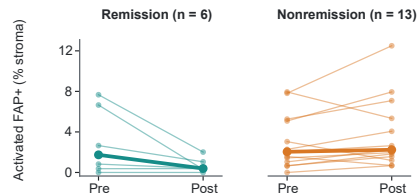
B FAP+ fibroblast adhesion state

16 patients; median paired Δ = 0.07



D Activated FAP+ cells after anti-TNF

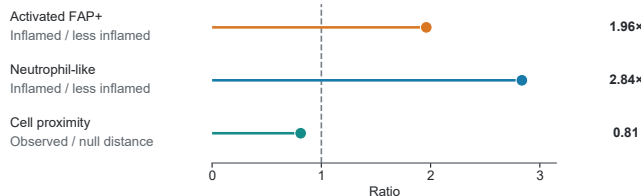
Site-matched pre/post adalimumab



Lines: patients; bold lines: group medians

E Paired spatial validation in UC

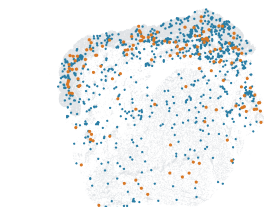
One donor; paired UC sections (SCP3818)



Proximity: P = 0.005; 1,000 size-matched null sets

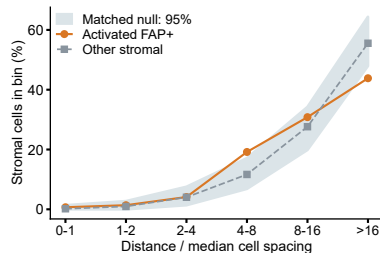
F Spatial context of the activated-FAP niche

UC1 inflamed; one donor



● Neutrophil-like (n=879)
● Activated FAP+ (n=146)

Nearest neutrophil-like cell



G Cross-system pathway responses

